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A memetic algorithm based on two_Arch2 for multi-depot heterogeneous-vehicle capacitated arc routing problem

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ABSTRACT

With the rapid growth in the number of motor vehicles, traffic pollution has become an increasingly serious problem, due to high carbon emission and low load utilization rate. It is reasonable to plan vehicle driving routes on urban roads to deliver services, (i.e., to tackle the capacitated arc routing problem (CARP)). Previous studies on CARP have typically considered at most two objectives simultaneously; however, two objectives are not sufficient to solve the pollution problem. Consequently, a many-objective optimization model of multi-depot heterogeneous-vehicle CARP is constructed in this study that considers four objective functions: total cost, makespan, carbon emission and load utilization rate. Furthermore, based on the ideas of two-archive and information entropy, we propose a memetic algorithm based on Two_Arch2 (MATA) to tackle the constructed model. The experimental results show that MATA effectively optimizes the many-objective model and solves the problem.

1. Introduction

The capacitated arc routing problem (CARP) is a special vehicle routing problem (VRP) [1] widely encountered in real life scenarios such as garbage cleaning, street watering, and network monitoring service [2,3]. The vehicle driving routes need to be adequately planned to meet individual circumstances to solve the CARP. As a typical NP-hard problem [4], the difficulty of tackling the CARP increases rapidly as the problem scale and the number of constraints increase. In a single-objective CARP, only one objective function needs to be considered, namely, finding the routes with the minimum total cost for all tasks under the constraint conditions [5]. In multiobjective CARP, the total cost and makespan are considered simultaneously [6]. However, one problem with the existing methods is that they cannot find an optimal solution in polynomial time [7]. In fact, a large number of practical applications have shown that using bio-intelligent algorithms to solve the CARP could substantially reduce global transportation costs [8].

As a classic NP-hard problem, CARP has been studied by many scholars for several decades, but most researches pay attention to the single-objective CARP or the multiobjective CARP. Heuristic algorithms [9] are widely used to solve such problems. Golden et al. [10] proposed the CARP, established the mathematical model, and provided the corresponding solution. Subsequently, many scholars have studied the CARP. Lacomme et al. [11] proposed a multiobjective CARP model that included two objectives—total cost and construction period. The memetic algorithm (MA) [12,13] has been proved to be an effective method to solve such problem. Pia et al. [14] proposed a model that combined both large and small vehicles; the small vehicles periodically transfer their goods into large vehicles and then continue to work. Then, they proposed a local search heuristic obtained from a variable neighborhood procedure to solve the CARP. Many scholars believe that continued worsening of the global greenhouse effect makes controlling the carbon emission of vehicles increasingly important. Bekta et al. [15] studied the pollution-routing problem (PRP) and proposed a method for calculating carbon emission. Tang et al. [16] proposed applying an MA to an extended neighborhood search to tackle the CARP. Mei et al. [6] proposed a decomposition-based MA (D-MAENS) for multiobjective CARP, which combined the MAENS method of single-objective CARP with the advanced functions of multiobjective optimization. And they adopted the cooperative co-evolution (CC) framework to decompose a large-scale CARP into smaller problems and tackled them separately [17]. They developed an effective decomposition scheme called route distance grouping (RDG). The CC mechanism is effective for improving the efficiency of optimization process [18]. Following the work of Mei et al., Shang et al. [19] proposed an improved MA based on path distance grouping to solve a multiobjective large-scale CARP (LSCARP). Recently, Yu et al. [20] studied a split-delivery mixed CARP and provided a mathematical formulation to accelerate the meta-heuristic algorithm search process.

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The traditional studies on the CARP only consider multiobjective multi-depot problems or multiobjective heterogeneous-vehicle problems. However, they have not combined these two aspects. The multi-depot [21] scenario refers to a traffic network with multiple depots in which every vehicle departs from and returns to a depot after completing its tasks or reaching an upper capacity limit. The heterogeneous vehicle [22] scenario involves various types of vehicles that feature different maximum capacities, speeds and other characteristics. In the real world, the practical engineering problems mostly involve many-objective problems with multiple depots and various kinds of vehicles.

In addition, most of the previous studies have used the MA to tackle the CARP, but in many-objective CARP, as the number of objectives increases, the traditional MA approaches degrade. Therefore, based on MA, this paper adopts the two-archive strategy [23,24] and combines local optimization with global optimization, which effectively prevents the algorithm from falling into local optima. Meanwhile, by introducing the information entropy strategy, this paper designs a dynamic crossover operator and a dynamic local search operator that more closely emulate the biological evolution.

Besides, via taking the carbon emission [25] and load utilization into account, this paper constructs a many-objective optimization model for multi-depot heterogeneous-vehicle CARP that reduces the total cost, makespan, and carbon emission while improving the load utilization rate. The main contributions of this paper are as follows:

1) We construct a many-objective optimization model with four objectives (total cost, makespan, carbon emission, and load utilization rate) to tackle the multi-depot and heterogeneous-vehicle CARP.

2) By combining the Two_Arch2 algorithm with the MA, we propose an MA based on Two_Arch2 (MATA) to optimize the constructed model.

3) We introduce the information entropy strategy, accordingly, the local search operator and the crossover operator are adjusted dynamically to adapt to different evolutionary periods.

2. Problem description and model construction

2.1. Basic CARP

The CARP is constructed on the basis of a weighted connected graph $G(V, E, A)$, where V represents a set of vertices, E represents a set of edges, and A represents a set of arcs. Each edge $(v_i, v_j) \in E$ and arc$< v_i, v_j >\in A$ in graph G includes the following attributes: the cost of serving the streets $serve\_cost(v_i, v_j) \geq 0$, the cost of travelling the streets $trav\_cost(v_i, v_j) > 0$, and the task demand $demand(v_i, v_j) \geq 0$. There is a specific point in the vertex set V called the depot. In addition, the maximum vehicle load is Q. The basic CARP involves designing a set of routes in graph G that satisfy the following constraints:

1) Each vehicle must start from a depot and finally return to the depot;

2) Each street can be passed several times, but can be served only once;

3) For each route, the load of the vehicle cannot exceed the maximum load Q.

Using the point coding method in [26], the mathematical model of the basic CARP with one vehicle is expressed as follows:

$$\begin{aligned} \min f_1 = & \sum_{k=1}^{L-1} \text{serve\_ cost}\left(v_k, v_{k+1}\right) \times x_k \\ & + \sum_{k=1}^{L-1} \text{trav\_ cost}\left(v_k, v_{k+1}\right) \times y_k \\ \text{s.t. }: & \sum_{k=1}^{L-1} \text{demand}\left(v_k, v_{k+1}\right) \times x_k \leq \text{Q} \\ & \left(v_k, v_{k+1}\right) \neq \left(v_j, v_{j+1}\right), \forall x_k = 1, x_j = 1, k \neq j \end{aligned} \quad (1)$$

where $x_k = 1$ or 0, denoting whether a vehicle services a street (v_k, v_{k+1}), y_k=1 or 0, denoting whether a vehicle travels through the street (v_k, v_{k+1}), and L is the total length of the street. The total cost includes two aspects: $serve\_cost(v_i, v_j)$ and $trav\_cost(v_i, v_j)$. Constraints ensure that the total load cannot exceed an upper limit and that each task will be served only once.

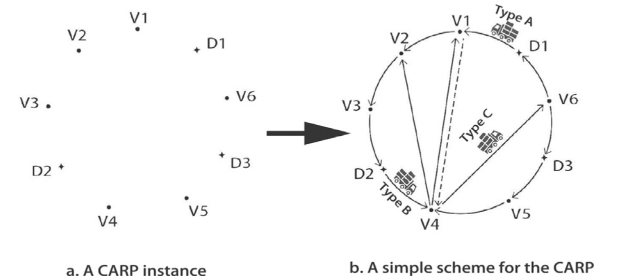

Fig. 1. The model of multi-depot heterogeneous-vehicle CARP.

2.2. Many-objective model of multi-depot heterogeneous-vehicle CARP

By combining many-objective optimization theory [27,28], carbon emission theory, load utilization rate [29], and multi-depot heterogeneous-vehicle CARP, this paper proposes a many-objective optimization model for multi-depot heterogeneous-vehicle CARP, which is described in detail as below.

Assume that a city has multiple depots; each depot has multiple vehicles; and that these vehicles are not of the same type and serve the edges or arcs of the city.

Fig. 1 shows a schematic diagram of urban garbage collection. The solid lines and dotted lines indicate routes requiring service or not, respectively. During the garbage collection process, the constraints are the same as those in the basic CARP. Four objectives must be optimized: total cost, makespan, carbon emission and load utilization rate. And the goal is to reduce the total cost, makespan, and carbon emission as much as possible, as well as increase the load utilization rate.

We denote $STR(ID, L, q)$ as the street information, $Y(ID, H, v)$ as the depot information, and $V\left(ID, h, y, C_1, C_2, C_3, Q_{\max}, Q, \rho_1(Q), \rho_2(Q)\right)$ as vehicle information.

The basic notations used in this section are as follows:

S: The number of streets in the city;
D: The number of depots in the city;
V_d^h: The number of h-type vehicles in the depot;
H_d: The number of vehicle types in the depot;
C_1^h: The fixed cost of an h-type vehicle;
C_2^h: The service cost per meter of an h-type vehicle;
C_3^h: The traversal cost of an h-type vehicle;
C_4: The environmental cost per kilogram of CO_2 emission;
q_i: The number of the tasks on street i;
L_i: The length of street i;
Q_{max}^h: The maximum capacity of an h-type vehicle;
$v_1^h(S_i)$: The speed of an h-type vehicle serving street i;
$v_2^h(S_i)$: The speed of an h-type vehicle traversing street i ;
$\rho_1^h(Q_i)$: The fuel consumption of an h-type vehicle in service of street i under the condition of Q_i load;
$\rho_2^h(Q_i)$: The fuel consumption of an h-type vehicle when travelling through street i under the condition of Q_i load;
ω: The CO_2 emission factor;
Q_i: The weight of a vehicle traveling on street i;

$$x_i^{dhv} = \begin{cases} 1 & h\text{ - type vehicle } v \text{ of depot } d \text{ serves street } i \\ 0 & \text{else} \end{cases}$$

$$y_i^{dhv} = \begin{cases} 1 & h\text{ - type vehicle } v \text{ of depot } d \text{ travels through street } i \\ 0 & \text{else} \end{cases}$$

2.2.1. Objective functions

A. Economic Cost: Economic cost [30] includes three parts: fixed costs, street serving costs and street traversing costs. The fixed costs include vehicle purchase and maintenance, which are related to the numbers and types of vehicles. The street serving and street traversing costs include the costs incurred during the garbage collection process. These costs are related to the length of street, the number of vehicles and the vehicle types. The economic cost function is as follows:

$$\min f_1 = \sum_{d=1}^{D}\sum_{h=1}^{H_d} V_d^h + \sum_{i=1}^{S}\sum_{d=1}^{D}\sum_{h=1}^{H_d}\sum_{v=1}^{V_d^h} C_2^h x_i^{dhv} L_i + \sum_{i=1}^{S}\sum_{d=1}^{D}\sum_{h=1}^{H_d}\sum_{v=1}^{V_d^h} C_3^h y_i^{dhv} L_i \tag{2}$$

B. Makespan: In practical garbage collection applications [31], for aesthetic reasons, municipal departments wish to complete garbage collection tasks as fast as possible. In [32], the French city of Troy starts garbage trucks on their routes at the same time every morning in an attempt to complete the entire garbage collection process as soon as possible so that the cleaning personnel can be assigned to other tasks. The makespan function is as follows:

$$\min f_2 = \max_{d\in D}\left(\max_{v\in\sum_{h=1}^{H} v_d^h, h\in H_d}\left(\sum_{i=1}^{S}\frac{x_i^{dhv}L_i}{v_1^h(S_i)} + \sum_{i=1}^{S}\frac{y_i^{dhv}L_i}{v_2^h(S_i)}\right)\right) \tag{3}$$

C. Carbon Emission: Different from the VRP, the carbon emission [33] of vehicles on arc paths include two factors: vehicle emission when servicing the street and vehicle emission when traversing the street. Carbon emission is related to driving distance and fuel consumption coefficient. The carbon emission function is as follows:

$$\min f_3 = C_4\omega\sum_{i=1}^{S}\sum_{d=1}^{D}\sum_{h=1}^{H_d}\sum_{v=1}^{V_d^h} x_i^{dhv} L_i \rho_1^h(Q_i) + C_4\omega\sum_{i=1}^{S}\sum_{d=1}^{D}\sum_{h=1}^{H_d}\sum_{v=1}^{V_d^h} y_i^{yhv} L_i \rho_2^h(Q_i) \tag{4}$$

D. Load Utilization Rate: The load utilization rate reflects the loading efficiency of the vehicles. It cannot be guaranteed that the vehicles will reach their maximum load capacities after completing their tasks; therefore, the load utilization rate can be only as close to the threshold value as possible. We use the relationship between the total lengths of the service routes and the travelled routes to quantify the load utilization. $x_i L_i$ represents the served street lengths for each vehicle, while $y_i L_i$ is the traversed street lengths for each vehicle. :The load utilization rate function is as follows:

$$\max f_4 = \frac{\sum_{i=1}^{S}\sum_{d=1}^{D}\sum_{h=1}^{H_d}\sum_{v=1}^{V_d^h} x_i^{dhv} L_i}{\sum_{i=1}^{S}\sum_{d=1}^{D}\sum_{h=1}^{H_d}\sum_{v=1}^{V_a^h}\left(x_i^{dhv} L_i + y_i^{dhv} L_i\right)} \tag{5}$$

2.2.2. Constraints

We assume that the load of an h-type vehicle cannot exceed its maximum capacity. The formula is as follows:

$$\sum_{i=1}^{S} x_i^{dhk} q_i \le Q_{\max}^h, \forall d,h,k \tag{6}$$

We also assume that each street must be serviced only once. The equation is as follows:

$$\sum_{d=1}^{D}\sum_{h=1}^{H_d}\sum_{v=1}^{V_d^h} x_i^{dhv} = 1, \forall i \tag{7}$$

Another assumption is that each street can be traversed many times. The formula is as follows:

$$\sum_{d=1}^{D}\sum_{h=1}^{H_d}\sum_{v=1}^{V_d^h} y_i^{dhv} \ge 0, \forall i \tag{8}$$

Finally, we assume that each vehicle starting from a depot will return to its original depot. The equation is as follows:

$$y_0^{dhv} = y_N^{dhv}, \forall d,h,v \tag{9}$$

3. Algorithm design

3.1. Definition of many-objective optimization

In contrast to single-objective problems, the solutions of many-objective problems must optimize many conflicting objectives simultaneously [34]. For an NP-hard combination problem, heuristic algorithms can generate a set of non-dominated solutions. Therefore, we adopted heuristic algorithms to solve the many-objective CARP in this paper. The mathematical form of multiobjective optimization is generally defined as [35,36]:

$$\min \vec{f}(\vec{x}) = \left\{f_1(\vec{x}), f_2(\vec{x}), \dots f_n(\vec{x})\right\} \tag{10}$$

s.t.

$$\begin{cases} g_i(\vec{x}) \le 0 & i = 1,2,\dots m \\ h_i(\vec{x}) = 0 & i = 1,2,\dots p \end{cases} \tag{11}$$

where $\vec{x} = (x_1, x_2, \dots x_n)^T$ is a vector composed of decision variables, $f_i(\vec{x})$ is the i-th objective function, and $g_i(\vec{x})$ and $h_i(\vec{x})$ are the constraint conditions. In particular, when $n > 3$, a multiobjective optimization problem is called a many-objective optimization problem. The ultimate goal is to obtain a set of Pareto-optimal solutions that are as close to the Pareto front (PF) as possible [37] while guaranteeing the convergence and distribution of the solution set insofar as possible.

3.2. Dynamic adjustment of depots and heterogeneous-vehicle

For the closed multi-depot problem, according to the distance priority principle, task arcs are assigned to the nearest depots, which transforms the multi-depot problem into several single-depot problems. Then, the depot information is determined based on the boundary arc dynamic adjustment strategy [3].

For heterogeneous-vehicle task allocation problems [38], we must consider not only the path selection but also the vehicle selection. Arrangements of different types of vehicles can have an enormous effect on the results. In this paper, the vehicles are classified based on the paths. To avoid having large-capacity vehicles serving small-return roads, this paper sets a threshold value α, as shown in Eq. (12). When the ratio of circuit demand to the maximum vehicle capacity is greater than α, these vehicles are called same-group vehicles (SGV). For a certain route R_i, these constraints are met by exchanging vehicles among the $SGV(R_i)$. Then, using Eqs. (13) and (14), we can find an optimal set of vehicles and depots that achieving the maximum fitness value. The specific steps are detailed in Algorithm 1.

$$\delta = \frac{Q(R_i)}{Q_{\max}^j} \tag{12}$$

$$fit = \sum_{i=1}^{N}\lambda_i f_i' \tag{13}$$

$$f_i' = \frac{f_i - \min_i}{\max_i - \min_i} \tag{14}$$

3.3. Information entropy strategy

The information entropy represents the uncertainty inherent in information and can be used to describe the probability of information [39]. Information entropy in a population can be expressed as follows:

$$E = -\sum_{i=1}^{N} p_i \log|q_i| \tag{15}$$

Algorithm 1: SBX_Vehicle

Input: loop path number: R_i;the total number of loop path: N;demand of loop path i: $Q(R_i)$;the maximum capacity of h-type vehicles: $Q^h_{\max}$;the number of types: H^d;Crossover probability: P.
Output: The chosen vehicle type for R_i: best (R_i).

$SGV(R_i) \leftarrow \emptyset$;
for $R_i = 1$ *to* N **do**
  for $j = 1$ *to* H^d **do**
    if $Q(R_i) \leq Q^j_{\max}$ *and* $\delta > \alpha$ **then**
      $SGV(R_i) \leftarrow SGV(R_i) \cup \{V^j\}$;
    else
      break ;
for $i = 1$ *to* $SGV(R_i)$ **do**
  if $P < rand$ **then**
    find the best $fit(V^i)$;
    best $(R_i) \leftarrow V^i$;

where $p_i = \frac{|q_i|}{M}$, M is the number of individuals in the population, $|q_i|$ is the number of individuals included in q_i, and N is the number of individuals in the same range in the population after dividing the objective values into M equal parts in the feasible solution range.

By combining the many-objective optimization problem with the information entropy theory, during the many-objective optimization process, the probabilities of crossover and local search during population evolution are as follows:

$$\begin{cases} p'_c = \alpha_1 - \frac{\alpha_1 N}{M} \\ p'_l = \alpha_2 - \frac{\alpha_2 N}{M} \end{cases} \tag{16}$$

where $\alpha_1, \alpha_2 \in (0, 1)$ are parameters. As the population evolves, the smaller N is, the greater the probability of crossover and local search, and the better the diversity of the population. Conversely, the larger the N is, the smaller the probabilities of crossover and local search, and the better the convergence of the population. Thus, introducing the information entropy mechanism helps to better simulate the process of population evolution.

The population evolution process can be divided into three stages: the initial stage, the intermediate stage and the final stage. During the initial population evolution stage, the population diversity is rich and individual similarity is low. To accelerate population evolution, the population crossover probability can be increased, and the local search probability can be reduced. During the second stage of population evolution, to prevent local optimization, the local search probability should be increased accordingly, and during the last stage of evolution, the population structure is relatively simple; therefore, to accelerate the algorithm, the crossover probability and local search probability should be reduced. According to the above analysis, the crossover probability follows a decreasing function model during the evolution, and we adopt the Newton cooling model. In contrast, the local search probability follows a function model that first increases and then decreases; therefore we adopt the Gaussian function model. These two formulas are as follows:

$$\begin{cases} p''_c = p''_c(k_0) \times e^{-tk} \\ p''_l = ae^{-\frac{(k-b)^2}{2c^2}} \end{cases} \tag{17}$$

where k is the generation number, k_0 denotes the first generation, $p''_c(k_0)$ is the crossover probability in the first generation, and t, a, b and c are real constants with t and a greater than 0.

In conclusion, the final crossover probability and local search probability are given by:

$$\begin{cases} p_c = \alpha_1 - \frac{\alpha_1 N}{M} + p''_c(k_0) \times e^{-tk} \\ p_l = \alpha_2 - \frac{\alpha_2 N}{M} + ae^{-\frac{(k-b)^2}{2c^2}} \end{cases} \tag{18}$$

3.4. Memetic algorithm based on two_Arch2

The MA is an intelligent algorithm that combines the genetic algorithm (GA) [40] and a local search strategy. Via the mechanism of combining the local search strategy, the individuals in the population can achieve local optimization during each iteration. MA has high optimization efficiency and strong fault tolerance. Although the initial population may include individuals far from the optimal solution, the algorithm can also eliminate the individuals with poor fitness through the genetic operations and the local search strategy, causing the feasible solutions to constantly approach the optimal solution. However, when solving large-scale many-objective optimization problems, it is difficult to find an optimal solution using only the local search strategy.

In solving constrained many-objective optimization problems, it is important to consider both convergence and diversity. Based on the idea of two-archive, Two_Arch2 [41,42] maintains two archives simultaneously: a convergence archive (CA) and a diversity archive (DA). The CA induces individuals to move to the PF quickly, while the DA maintains a diverse population. Individuals can be subjected to crossover between CA and DA, but they mutate only in the CA (because crossover and mutation are carried out independently). The sizes of the CA and DA are fixed individually and they are updated using different strategies. The final DA is output as the final solution set.

Inspired by the idea of two archives, this paper proposes an MA based on Two_Arch2 (MATA) to tackle the many-objective CARP optimization problem. First, the population is initialized by the Dijkstra algorithm [43], which calculates the shortest distance between any two points. Second, two archives are constructed, the non-dominated solutions in the DA and CA are empty. Then, individuals are subjected to crossover, and depots and vehicles are adjusted between the CA and DA. Local search is performed only in the CA by ACO [44] (crossover and local search are independent). Then, the CA and DA are updated independently. Finally, take DA as the final output when the algorithm reaches the maximum number of iterations. The specific steps are listed in Algorithm 2.

Algorithm 2: MATA

Input: population: $G(V, E, A)$;crossover probability : $P_c(ite)$;local search probability : $P_l(ite)$;ACO related variables.
Output: DA

$POP \leftarrow$ initialize population using Dijkstra(v_i, v_j);$DA \leftarrow$ find non-dominated solution(POP), $CA \leftarrow \emptyset$;
for $ite = 1$ *to* G_{max} **do**
  if *rand* $< P_c$*(ite)* **then**
    $NPOP_1 \leftarrow$ SBX(CA,DA); $NPOP_2 \leftarrow$ SBX_vehicle($NPOP_1$)by Algorithm 1;
  if *rand* $< P_l$*(ite)* **then**
    $NPOP_3 \leftarrow$ ACO(DA);
  $NPOP \leftarrow (NPOP_2 \cup NPOP_3)$; $CPOP \leftarrow$ find non-dominated solution(NPOP); $[CA, DA] \leftarrow$ update(CPOP);

4. Experimental results and analysis

In this experiment, we compared MATA with MDILSMA/D [3], IACO [1] and D-MAENS [6] on the gdb set, the val set and the Chicago set, and evaluated the results with the hypervolume (HV) and the visualization.

Table 1
HV values of MATA, MDILSMA/D, IACO and D-MAENS on the gdb set.

Problem	MATA	D-MAENS	MDILSMA/D	IACO	winner
gdb1	0.1027	0.0383	0.0853	0.337	MATA
gdb2	0.0526	0.0415	0.0498	0.0377	MATA
gdb3	0.0536	0.0652	0.1050	0.0333	MDILSMA/D
gdb4	0.0612	0.0457	0.0612	0.0360	MATA and MDILSMA/D
gdb5	0.0527	0.0427	0.0898	0.0242	MDILSMA/D
gdb6	0.0562	0.0787	0.0760	0.0353	MDILSMA/D
gdb7	0.0583	0.0760	0.1005	0.0290	MDILSMA/D
gdb8	0.0698	0.0461	0.0579	0.0433	MATA
gdb9	0.0721	0.0410	0.0580	0.0365	MATA
gdb10	0.0507	0.0606	0.0732	0.0247	MDILSMA/D
gdb11	0.0572	0.0423	0.0569	0.0365	MATA
gdb12	0.0458	0.0413	0.0910	0.0265	MDILSMA/D
gdb13	0.0691	0.0430	0.0422	0.0609	MATA
gdb14	0.0506	0.0456	0.0458	0.0360	MATA
gdb15	0.0514	0.0304	0.0589	0.0226	MDILSMA/D
gdb16	0.0534	0.0320	0.0441	0.0285	MATA
gdb17	0.0590	0.0345	0.0555	0.0255	MATA
gdb18	0.0533	0.0453	0.0435	0.0267	MATA
gdb19	0.0797	0.0340	0.0393	0.0504	MATA
gdb20	0.0591	0.0502	0.0586	0.0253	MATA
gdb21	0.0524	0.0203	0.0458	0.0206	MATA
gdb22	0.0554	0.0193	0.0454	0.0179	MATA
gdb23	0.0403	0.0292	0.0524	0.0320	MDILSMA/D

4.1. Hypervolume (HV)

To evaluate the set of obtained non-dominanted solutions, we should consider two aspects: the convergence and diversity of the solution set, that is, the solutions should be as close to the PF as possible, and the distribution of the non-dominated solution set, which should be evenly distributed along the ideal PF. Therefore, two commonly used performance indicators are selected for this study: the hypervolume (HV) [24] and the performance visualization indicator. The HV represents the volume of the region formed by the solutions and the reference point in the objective space. The formula is as follows:

$$HV(P,Z) = \text{volume}\left(\bigcup_{p\in P} \left[p_1, z_1\right] \times \dots \left[p_N, z_N\right]\right) \tag{19}$$

where P is the obtained set of non-dominanted solutions and Z is the reference point. A larger HV value indicates better convergence and distribution.

4.2. Visualization

In many-objective optimization problems, it is nearly impossible to observe the evolution process in the many-objective space directly because it is difficult to describe the high-dimensional hyperplane. He et al. [45] proposed a new visualization method that maps the non-dominated solution set from the high-dimensional objective space to a two-dimensional polar graph. This approach preserves both the Pareto dominance relationships and the shape and position of the PF. Using this method, the diversity and convergence of solutions can be observed intuitively.

4.3. Algorithm comparisons

Table 1 and Table 2 show the HV values of MATA, MDILSMA/D, IACO and D-MAENS on the gdb and val sets. As Table 1 shows, MATA achieved the best performance, winning 15 times, while MDILSMA/D was second best, winning 9 times on the gdb set. In Table 2, MATA again obtained the best performance, winning 9 times, while MDILSMA/D performed second best, winning once on the val set. These results show that the MATA algorithm is stable on these test sets.

Table 3 shows the HV values and objective function values obtained by MATA, MDILSMA/D, IACO and D-MAENS on the Chicago example.

Table 2
HV values of MATA, MDILSMA/D, IACO and D-MAENS on the val set.

Problem	MATA	D-MAENS	MDILSMA/D	IACO	winner
val1	0.0899	0.0246	0.0975	0.0358	MDILSMA/D
val2	0.0790	0.0135	0.0640	0.0586	MATA
val3	0.0612	0.0619	0.0799	0.0230	MATA
val4	0.0433	0.0081	0.0214	0.0182	MATA
val5	0.0506	0.0054	0.0209	0.0214	MATA
val6	0.0597	0.0373	0.0139	0.0228	MATA
val7	0.0532	0.0164	0.0488	0.0229	MATA
val8	0.0501	0.0122	0.0125	0.0240	MATA
val9	0.0408	0.0095	0.0365	0.0152	MATA
val10	0.0426	0.0064	0.0238	0.0144	MATA

Table 3
HV values and objective-function-optimized results of MATA, MDILSMA/D, IACO and D-MAENS on the Chicago set.

Algorithm	HV	f_1/Yuan	f_2/minute	f_3/Yuan	f_4
MATA	**0.0230**	2154.766	57.254	779.952	92.897%
D-MAENS	0.0062	3073.363	50.928	971.719	82.872%
MDILSMA/D	0.0223	2288.199	47.079	573.151	91.015%
IACO	0.0208	2333.683	55.056	632.853	91.294%

Table 3 also shows that MATA achieved the best HV value (0.0230). The function values for f_1, f_2, f_3 and f_4 were obtained by averaging the non-dominated solutions and are also listed in Table 3. As the table shows, after optimization, MATA achieved the best economic cost f_1 and load utilization rate f_4. Therefore, MATA can better optimize this complex CARP.

Fig. 2 shows the visualizations of the four algorithms. There are some problems in creating the visualizations because the mutation operator is replaced by local search in MA. In addition, the traditional MA cannot effectively deal with many-objective optimization problems. The visualized performance of MATA remains good.

5. Conclusion

In this paper, we designed and tested a many-objective model, with four objectives, economic cost, makespan, carbon emission and load utilization, to solve the multi-depot and heterogeneous-vehicle CARP problem. Based on the model characteristics, we designed MATA to optimize the driving routes of urban garbage collection vehicles.

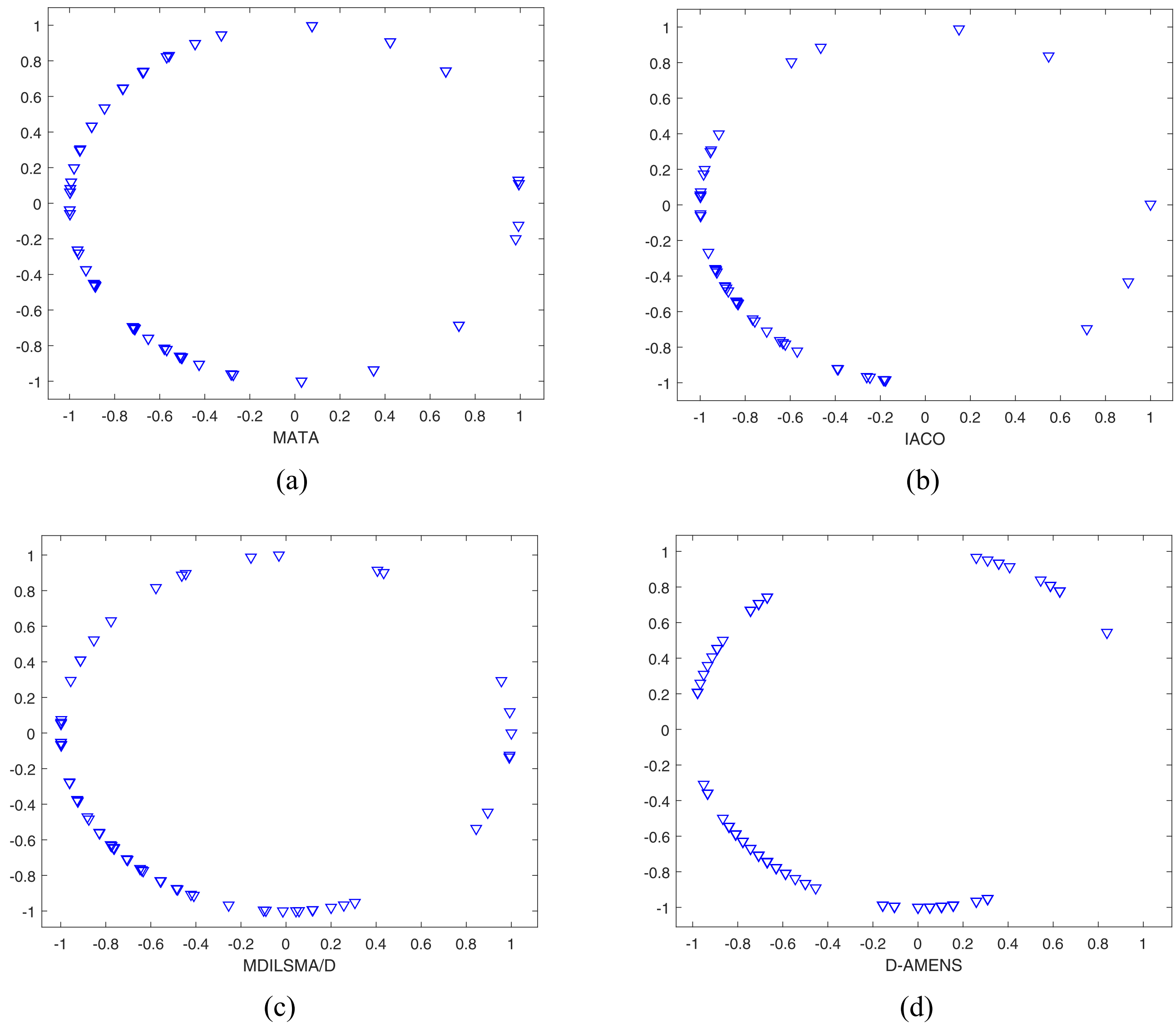

Fig. 2. Visualizations of MATA, IACO, MDILSMA/D and D-MENAS.

The results of the simulation experiments in this paper comparing MATA with MDILSMA/D, IACO and D-MAENS show that the MATA is able to solve the many-objective optimization model proposed in this paper effectively. The goal of our future work is to find a series of PSs at different times by considering the time-window mechanism and the dynamic multiobjective operator.

Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

We believe that this manuscript is appropriate for publication in the Swarm and Evolutionary Computation. This manuscript has not been published and is not under consideration for publication elsewhere. All authors have read and approved the final version of the manuscript.

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